

20 Credit Project Report

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Links

Project Repository: <https://github.com/GVCL/IWSeg-SAR-Poison.git>

Results Repository before the project start: https://iiitbac-my.sharepoint.com/:f:/r/personal/siddharth_kothari_iiitb_ac_in/Documents/IWSeg-SAR-Poison?csf=1&web=1&e=7VDsDU

Numeric Metric Results: <https://docs.google.com/spreadsheets/d/1wkcRR2JQirzWbJMN4lpj4b4C0Y0zzUewHl608aBp5aM/edit?usp=sharing>

Review Comments Doc: https://docs.google.com/document/d/luNmE4IcmS1TZ_aXcd2HEVRhVQCO9tp61kiqEH_mSD94/edit?usp=drivesdk

20 credit backup: https://iiitbac-my.sharepoint.com/:f:/g/personal/siddharth_kothari_iiitb_ac_in/Eh2f_2GwOdZLqxsXD-8V2EMBds__DvNcuvG0PfXMiSSiAQ?e=nweaHc

Dataset Figshare Link [1]: https://figshare.com/articles/dataset/Adversarial_Attack_Erosion_and_Dilation_Dataset_for_Amazon_River_Basin/28784405?file=53625587

Previous version arXiv preprint link [2]: <https://arxiv.org/abs/2505.01884>

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Table 1: Performance of the DeepLabV3 model on our dataset at various corruption levels

Corruption	Dice Coeff.	IoU	Precision	Recall	F1 Score	Specificity	Accuracy
0%	0.7699	0.8195	0.8913	0.8653	0.8645	0.9655	0.9533
2%	0.7579	0.8066	0.8787	0.8599	0.8538	0.9544	0.9443
12%	0.7123	0.7740	0.8778	0.8252	0.8248	0.9458	0.9332
15%	0.7055	0.7561	0.8508	0.8299	0.8161	0.9301	0.9215
17%	0.7225	0.7744	0.8662	0.8333	0.8267	0.9370	0.9286
20%	0.6779	0.7427	0.8586	0.8028	0.8010	0.9379	0.9193
25%	0.7124	0.7637	0.8568	0.8287	0.8224	0.9473	0.9244
30%	0.6708	0.7363	0.8576	0.8001	0.7981	0.9547	0.9179

Table 2: Performance of the SegFormer model on our dataset at various corruption levels

Corruption	Dice Coeff.	IoU	Precision	Recall	F1 Score	Specificity	Accuracy
0%	0.7770	0.8291	0.9079	0.8678	0.8701	0.9746	0.9566
2%	0.7716	0.8196	0.8893	0.8699	0.8630	0.9551	0.9503
12%	0.6411	0.7281	0.8275	0.7998	0.7876	0.9439	0.9241
15%	0.6464	0.7219	0.8577	0.7912	0.7875	0.9427	0.9183
17%	0.6401	0.7173	0.8573	0.7831	0.7830	0.9492	0.9161
20%	0.6332	0.7077	0.8136	0.7974	0.7763	0.9303	0.9063
25%	0.6182	0.6937	0.8259	0.7778	0.7604	0.9251	0.8947
30%	0.5972	0.6769	0.8060	0.7782	0.7502	0.9114	0.8898

1 Pre Midterm

1.1 Documentation of Reviewer Comments and Developing a Plan to Address them

The first step that we undertook was to read each reviewer comment provided for the JSTARS version of the paper in detail. We filtered out and sorted each of the review comments, based on whether we feel it is relevant to our line of work, or if it is out of scope of the paper, and grouped similar recommendations (such as ones requiring additional experimentation, or ones requiring paper rewriting). We then developed a plan to address each of them in detail, along with the steps required, and how they will tie in into the new version. The broad steps include -

1. Additional Experiments with all the different models for the various levels of corruption, and use it to provide a comparative analysis, and experiments to validate our method as a simulation of manual annotation errors.
2. Paper Rewriting to make a narrative around each of the additional experiments, and to include their results. Apart from that, rewriting the original manuscript to make it more coherent and structured.
3. Reading additional reference material to make our work more robust and scientifically grounded. This included references to justify why we chose Sentinel-1 images, a justification for using morphological operations etc.

1.2 Additional Experiments - Corruption Analysis for all models

We conducted additional experiments with the various corruption levels (2%, 12%, 15%, 17%, 20%, 25%, 30%) for each of the models (Segformer, Maskformer and DeepLabV3). We trained the models for each of these levels, and obtained evaluation metrics which are stated in Tables 1, 2 and 3. We also ran a qualitative analysis of the model outputs using a few examples which are shown in Figure 1. The results are discussed below.

Table 1 shows the evaluation metrics for the DeepLabV3 model, while Tables 2 and 3 show the results for SegFormer and Mask2Former, respectively. The results are along expected lines for DeepLabV3 and SegFormer, where we observe a gradual decay in performance with the increase in corruption levels. SegFormer has almost a perfect decay, while DeepLabV3 has a few variations, but also exhibits a gradual trend of decrease in performance.

Figure 1 (first 3 columns) shows the predictions of the DeepLabV3 model with the increase in the corruption levels. We observe that the sharp river boundaries seen in the ground truth and in the predicted mask for the uncorrupted case lose their sharpness with the increase in the corruption level. The model's predictions are

Table 3: Performance of the Mask2Former model on our dataset at various corruption levels

Corruption	Dice Coeff.	IoU	Precision	Recall	F1 Score	Specificity	Accuracy
0%	0.774	0.8269	0.8939	0.8721	0.8688	0.9727	0.9568
2%	0.7527	0.8038	0.8604	0.8681	0.8488	0.9556	0.9422
12%	0.7177	0.7928	0.8533	0.8441	0.8366	0.9656	0.9489
15%	0.7259	0.7964	0.8568	0.8469	0.8400	0.9639	0.9483
17%	0.7691	0.8201	0.8955	0.8642	0.8643	0.9738	0.9526
20%	0.7552	0.8088	0.8655	0.8718	0.8554	0.9519	0.9492
25%	0.7271	0.7992	0.8741	0.8414	0.8427	0.9721	0.9499
30%	0.7375	0.8032	0.8601	0.8639	0.8483	0.9621	0.9514

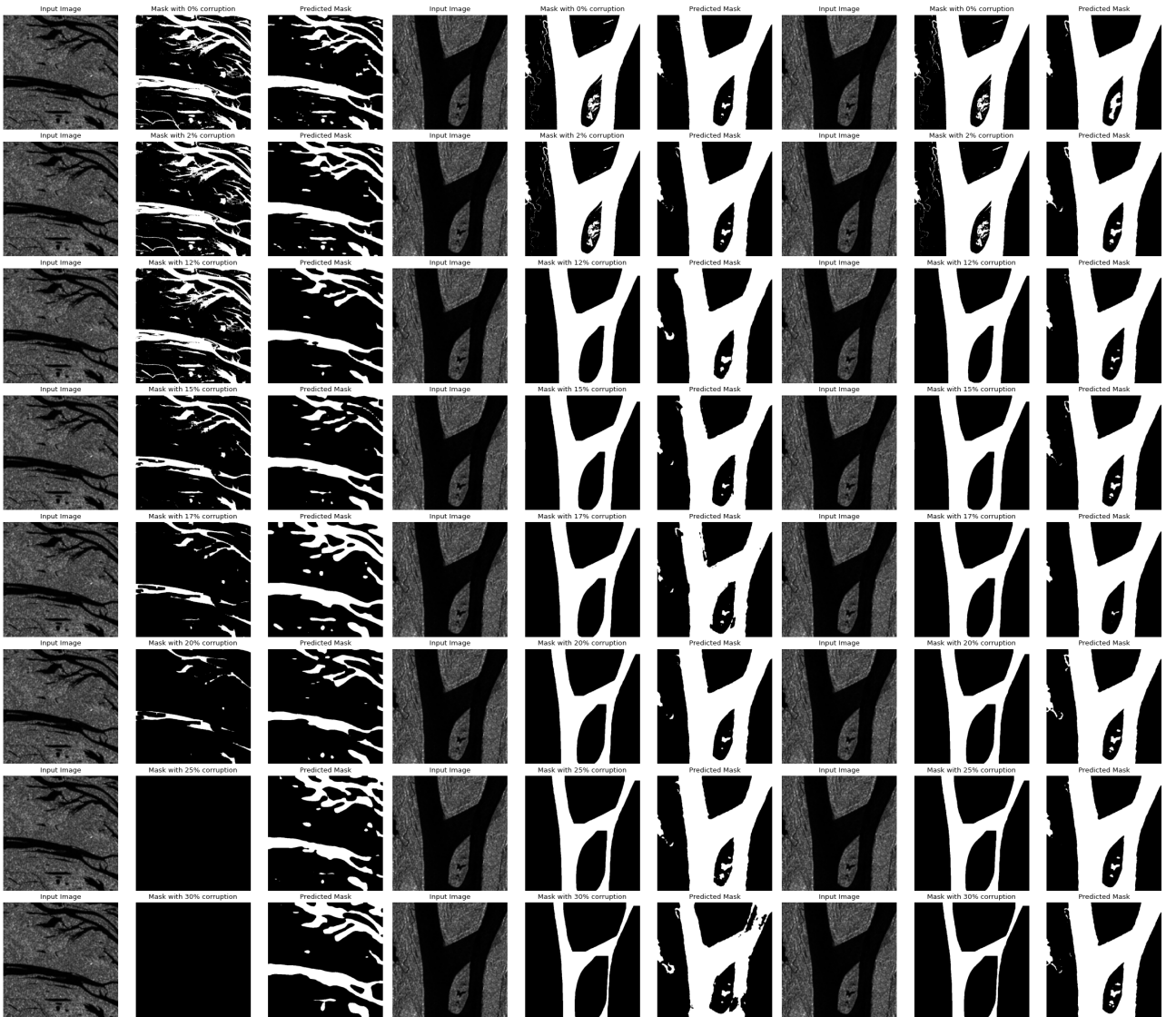


Figure 1: First Set of 3 columns - Results showing the effect of corruption level on the ground truth (second column) and the final predicted (rightmost column) mask by the DeepLabV3 model, for the input (leftmost column). The predicted masks lose sharpness in the model predictions with an increase in the corruption. Next Set of 3 columns - Segformer Model. Last Set of 3 columns - Maskformer Model.

sharp and well defined for the uncorrupted case, which later become increasingly smoother with corruption. Figure 1 also shows the same results for the SegFormer model. As observed, there is a constant decay in performance with the increase in corruption levels, which is consistent with the observed results.

The most interesting results are shown by Mask2Former, as shown in Table 3. The performance of the model never deteriorates and remains constant across all levels, with slight fluctuations. The lack of a proper trend indicates the fact that the dataset size is not large enough for Mask2Former to reliably learn. Instead, different runs of the model learn to varying degrees. This is also highlighted in Figure 1 (last 3 columns). As observed, there is almost no decay in performance. This indicates the need for larger training data for transformer-based models.

1.3 Additional Experiments - Manual Annotations and Simulation of Errors

We also generated masks for a few images via manual annotation. The goal of this experiment was to observe how errors occur during manual annotations, what are their types, and to provide a mathematical justification for using morphological operations to simulate these errors. The annotations were done using the CVAT [3] tool. The ground truths obtained from the NDWI method were then corrupted via various configurations of erosion and dilation to see whether these operations indeed model the manual annotation errors introduced.



Figure 2: 2 sets of images; For the leftmost 4 - Manually annotated mask using CVAT tool (top-right) as compared to ground truth mask (top-left) for a sample image, and eroded and dilated versions of the ground truth mask (bottom-left and right), respectively; For the rightmost 4 - Manually annotated mask using CVAT tool (top-right) as compared to ground truth mask (top-left) for a sample image. A combination of erosion and dilation is applied to this image, with the eroded region shown in the red box in the bottom-left (with dilation applied in the remaining image), and the final mask in the bottom-right.

In Figure 2 (the 4 images on the left), we apply erosion and dilation separately on the original NDWI mask, and compare the SSIM values of the three masks (NDWI Mask, Eroded Mask, and Dilated Mask) with the manually annotated masks using CVAT. The original image has an SSIM value of 0.726 when compared with the manually annotated mask, while the eroded and dilated masks have SSIM values of 0.739 and 0.708, respectively. Thus, in this case, erosion of the NDWI mask leads to a mask that is closer to the manual mask, which simulates the errors in the manually annotated mask. In Figure 2 (the 4 images on the right), we observe that the annotation errors can be best modeled by a combination of erosion and dilation (dilation primarily for the tributary and erosion for the islands formed). When only erosion and dilation are used, the SSIM values obtained upon comparing with the manually annotated mask are 0.875 and 0.873, respectively (compared to an SSIM value of 0.870 for the original NDWI mask), while the mask with partial erosion and dilation has an SSIM value of 0.877 when compared with the manually annotated mask.

1.4 Additional Paper Readings

The following is a summary of the papers we went through in order to strengthen our work. They were cited directly to strengthen our arguments on certain points or to provide more clarity.

1. An Analysis of the Impact of Annotation Errors on the Accuracy of Deep Learning for Cell Segmentation [4] - Shows that human boundary-drawing errors can be modeled using dilation and erosion, validating that these operations realistically simulate common annotation biases.
2. Flood detection from multi-temporal SAR data using harmonic analysis and change detection [5] - Sentinel-1 provides strong intensity contrast over water, making it more reliable for boundary detection than optical sensors like Sentinel-2.
3. Relationship of Local Incidence Angle with Satellite Radar Backscatter for Different Surface Conditions [6] - Low backscatter from surfaces like sand under certain angles can resemble water, showing that backscatter alone can be misleading and thresholding is often inadequate.
4. Noise Removal and Enhancement of Binary Images Using Morphological Operations [7] - Morphological operations are context-sensitive and non-linear, making them insufficient to reverse annotation errors, which supports using U-net as a denoiser.
5. An Optical and SAR Based Fusion Approach for Mapping Surface Water Dynamics over Mainland China [8] - High-frequency, accurate water monitoring aids climate resilience and adaptation strategies, highlighting the need for precise segmentation methods.
6. Optimal Fusion of Multispectral Optical and SAR Images for Flood Inundation Mapping through Explainable Deep Learning [9] - Combining Sentinel-1 and Sentinel-2 leverages SAR's all-weather reliability and optical spectral richness, improving segmentation accuracy and reducing misclassification.

1.5 Contributions to Paper Rewriting

A large section of the pre-midterm work was towards paper rewriting according to the review comments provided by the JSTARS reviewers. The edits are as follows -

1. The first rewriting of the paper was to make the paper more generalised. Earlier results were primarily targeted towards U-Net and its corruption analysis, which was changed drastically to accommodate results for each model. New qualitative results along with observations were added for each model, with the introduction being changed to suit the new objective; to provide a comparative analysis across models.
2. Information regarding the additional experiment for manual annotations was added. This included relevant figures, along with a discussion of the SSIM values observed and their implications.
3. The related work section was made more detailed and divided into four broad sections - focusing on Segmentation Work in remote sensing, existing datasets for SAR Segmentation, work pertaining to the simulation of manual errors and their types, and experiments in this field with Adversarial Attacks.
4. An important line of work which we pursued was pertaining to a review comment questioning why Sentinel-1 images were used when they can't distinguish between water and water like surfaces. This required a deep dive into the reasons for using Sentinel-1, and why they are combined with approaches involving Optical imagery, which provided a rationale for our methodology, and a justification for its usage.
5. We also added a section to talk about how U-Net can be used for morphological noise removal (and why just "reversing" it will not work). This section summarizes the primary result of U-Net efficiently tolerating morphological corruption upto high thresholds, and being ideal to deal with morphological noise.
6. Other aspects of rewriting included making each section more detailed, including adding more details about the parameters used for preprocessing, providing reasons to use the Amazon Basin, commenting in more detail on the inadequacy of previous datasets, and making captions more detailed.

1.6 2 Page Summary and Miscellaneous work

Each of the experiments required additional codes and configurations to be written, and we also performed additional work to use space efficiently on the computer -

1. Prepared a 2-page summary of our previous work by condensing a 10+ page paper—removed the related work, shortened dataset generation via pre-print references, limited results to key quantitative metrics and one edited image, and adjusted style files for the conference format.
2. Reconfigured the repository on the lab system and personal laptops, reinstalling conda environments and dependencies to enable remote work.
3. Modified deeplab, segformer, and maskformer codes for bulk training across configurations, added manual annotation experiments, SSIM calculations, and visualization code included in the paper.
4. Conducted regular code cleanups and backed up all results on OneDrive, including trained models, evaluation results, predicted images with visualizations, and the dataset.

2 Post Midterm

2.1 Additional Paper Readings - for the paper draft

The following is a summary of the papers we went through in order to strengthen our work. These were mainly focused on finding papers which talk about why the amazon basin is indeed a very diverse terrain and the findings on this region can be generalized to a lot of different terrains.

1. Amazon Hydrology From Space: Scientific Advances and Future Challenges [10] - Highlights the Amazon’s hydrological diversity—rivers, wetlands, floodplains, and seasonally inundated forests—creating challenging segmentation conditions. Validation here ensures robustness to extreme variability, with accurate segmentation holding global scientific value.
2. Geomorphological diversity of rivers in the Amazon Basin [11] - Describes meandering rivers, anabranching channels, floodplain lakes, and braided systems, showing the Amazon’s comprehensive fluvial structures. Reliable segmentation across these morphologies implies strong generalizability to other basins.

2.2 Initial Experiments pertaining to Uncertainty and Results obtained

The first set of experiments that we did was introducing MC Dropout into the existing UNet model. This would allow us to measure uncertainty by running multiple inferences and taking the variance of the same as uncertainty. The results are summarized below -

1. We observed that the variance values were very low. While they were consistent within an image (the highest values were observed at the boundaries, mid range values were observed in the water regions, and low uncertainty values for the land pixels), the magnitudes themselves varied across images. This is observed in figure 3. The values are not comparable across images. This was the first corrective step we took - to generate global heatmaps.
2. The next step was to generate global heatmaps by finding the max variance across images and scaling the images based on that. Two outputs are shown in figure 4. As can be seen, most figures appeared black, due to the very low uncertainty values. This implied we needed to experiment with other metrics, as this metric was not sufficient to quantify the uncertainty. The metric we chose was predictive entropy.
3. We then experimented with Predictive Entropy as the metric. This gave more consistent results, even when no explicit scaling was used. The results were along previous lines, with magnitudes being higher and useful, and hence subsequent experiment utilizes this metric. An example is shown in figure 5.
4. An additional validation experiment was run with the Coastline dataset [12] to validate the experiment on another dataset. Results were along expected lines, and this metric is chosen for future experiments.

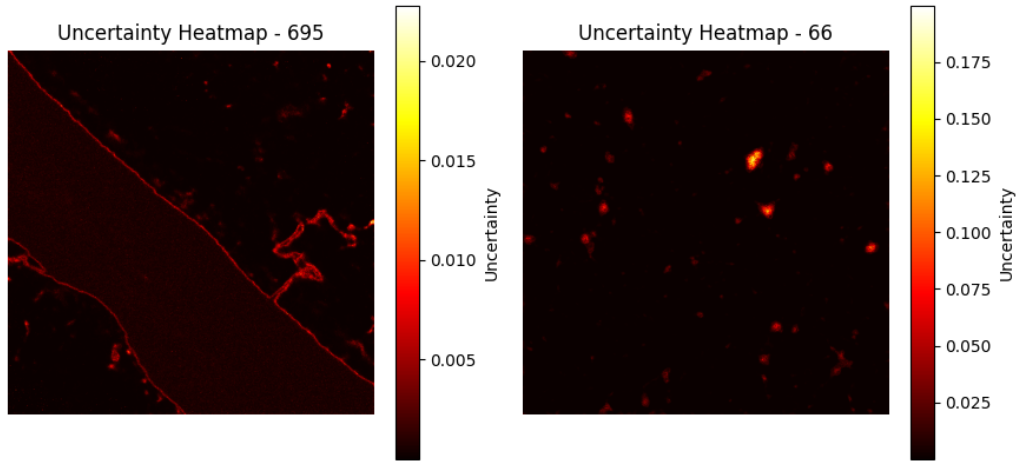


Figure 3: Unscaled Heatmaps for Variance Visualization

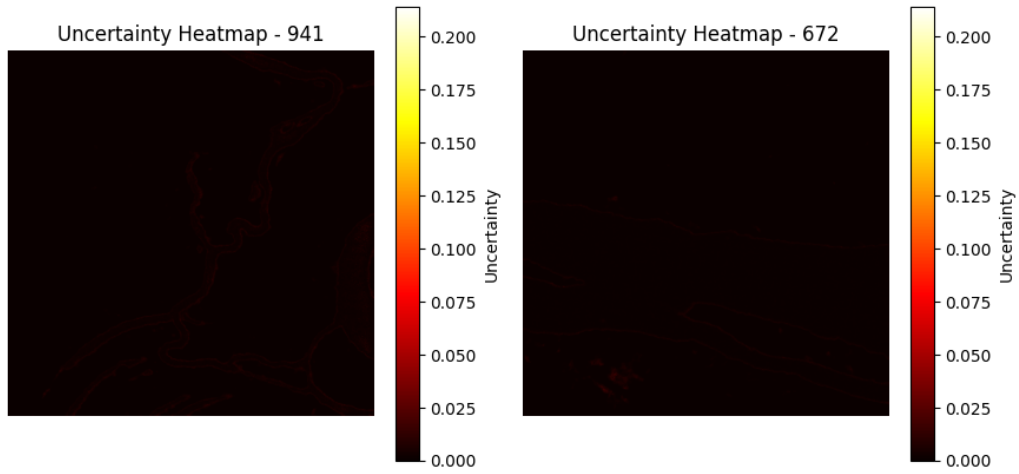


Figure 4: Scaled Heatmaps for Variance Visualization

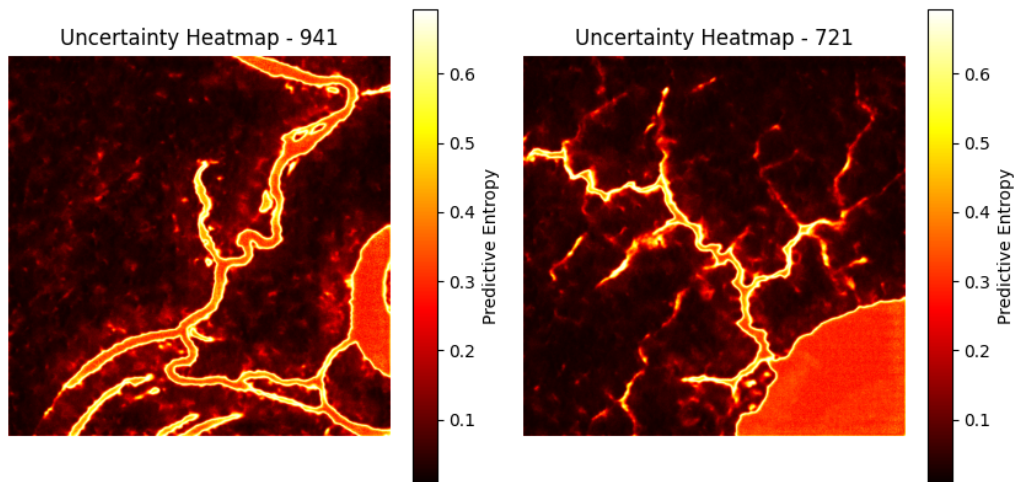


Figure 5: Entropy Heatmaps

2.3 Additional Paper Readings - for uncertainty mapping

We read a bunch of papers which introduce various methods to combat uncertainty, and also summarize existing methodologies. They are as follows -

1. Hyperbolic Uncertainty Aware Semantic Segmentation [13] - Introduces uncertainty-aware semantic segmentation in hyperbolic space, improving representation of hierarchical and structured relationships and enhancing robustness in ambiguous regions.
2. U-CE: Uncertainty-aware Cross-Entropy for Semantic Segmentation [14] - Proposes U-CE loss that incorporates pixel-wise predictive uncertainty via Monte Carlo Dropout, focusing training on uncertain and difficult regions.
3. Water-Land Segmentation via Structure-Aware CNN-Transformer Network on Large-Scale SAR Data [15] - Proposes Trans-SANet, a U-shaped CNN-Transformer hybrid capturing local spatial and global contextual features for water-land segmentation in SAR images.
4. Model Ensemble With Dropout for Uncertainty Estimation in Sea Ice Segmentation Using Sentinel-1 SAR [16] - Uses ensembles and dropout to highlight uncertainty in sea ice segmentation. However, ensembles were simple model repetitions without gating networks, simulating uncertainty via MC Dropout. We used the ResnetASPP implementation in later experiments.
5. UrbanSARFloods: Sentinel-1 SLC-Based Benchmark Dataset for Urban and Open-Area Flood Mapping [17] - Introduces the UrbanSARFloods dataset with annotated Sentinel-1 SLC imagery for urban and open-area flood mapping, useful for validation on unseen data.
6. Probabilistic SAR-Based Water Segmentation with Adapted Bayesian Convolutional Neural Network [18] - Compares Bayesian CNNs and Monte Carlo Dropout networks for water segmentation, showing BCNNs give more reliable uncertainty estimates while both achieve similar accuracy.
7. A Survey on Uncertainty Estimation in Deep Learning Classification Systems from a Bayesian Perspective [19] - Reviews uncertainty estimation techniques in deep learning, covering aleatoric and epistemic uncertainties, Bayesian networks, ensembles, MC dropout, and deterministic methods, helping identify strategies relevant to our work.

2.4 Ongoing Uncertainty Work and Results

Currently, the decided upon methodology which we are following involves the following steps -

1. We have generated 3 levels of corruption - low (2%), medium (8%) and high (15%). This is done using the previously available codes. We have also created 3 kinds of corruption - the previous one which is 1/3rd erosion, 1/3rd dilation and 1/3rd uncorrupted; completely erosion and completely dilation, as we want to observe the effects of erosion and dilation independently.
2. The models being tested are as follows - Dense U-Net, Residual U-Net (with residual blocks instead of Dense blocks), U-Net++ and ResnetASPP. The codes for U-Net++ are in development, while the others are ready and training.
3. One additional noise type which we are experimenting with is additive Gaussian white noise [20, 21, 22]. It is used to model real world noise in sensors and transmission, and is useful for robustness testing. We have created the code for Gaussian White Noise. We add the noise to the first band (VV band), while keeping the VH band untouched. The training of the models with this noise is now underway.

The results for the various metrics are shown in Tables 4 and 5 respectively. The trends show the following with regards to the accuracy and other metrics (outside of uncertainty) -

1. The models are able to largely withstand the mixed corruption, with minimal degrade in performance with the increase in corruption level. However, the decline is steeper as compared to the vanilla U-Net, which has no dropout layers.
2. The decline for erosion and dilation is steeper, and dilation completely breaks performance at the higher corruption level (this is also supported by the boxplots we had made earlier, which state that dilation constantly increases with the increase in corruption level, while erosion reaches a cap and stops increasing beyond a point).

Table 4: Performance of Dense Bayesian U-Net under different dataset configurations and corruption levels (Entropy-based evaluation)

Dataset Config	Corruption (%)	Metric						
		Dice Coeff	IoU	Precision	Recall	F1 Score	Specificity	Accuracy
Mixed	2	0.6887	0.7768	0.8264	0.8601	0.8252	0.9607	0.9470
	8	0.6597	0.7554	0.8204	0.8371	0.8056	0.9572	0.9356
	15	0.5826	0.7050	0.7948	0.7929	0.7617	0.9518	0.9211
Erosion	2	0.6782	0.7652	0.8649	0.8164	0.8132	0.9533	0.9303
	8	0.5785	0.6952	0.8262	0.7558	0.7577	0.9582	0.9118
	15	0.4919	0.6365	0.7702	0.7133	0.7070	0.9670	0.8915
Dilation	2	0.6624	0.7412	0.8005	0.8439	0.8028	0.9203	0.9253
	8	0.6065	0.6852	0.7536	0.8344	0.7636	0.8813	0.9013
	15	0.5418	0.6075	0.7104	0.8030	0.7062	0.8020	0.8353

Table 5: Performance of Residual Bayesian U-Net under different dataset configurations and corruption levels (Entropy-based evaluation)

Dataset Config	Corruption (%)	Metric						
		Dice Coeff	IoU	Precision	Recall	F1 Score	Specificity	Accuracy
Mixed	2	0.6736	0.7681	0.8351	0.8331	0.8142	0.9670	0.9431
	8	0.6163	0.7303	0.8147	0.8097	0.7819	0.9649	0.9302
	15	0.6623	0.7527	0.8221	0.8383	0.8074	0.9629	0.9340
Erosion	2	0.6926	0.7830	0.8542	0.8389	0.8279	0.9705	0.9487
	8	0.5963	0.7156	0.8248	0.7725	0.7705	0.9772	0.9345
	15	0.4355	0.6004	0.7666	0.6753	0.6740	0.9696	0.8780
Dilation	2	0.6678	0.7540	0.8072	0.8566	0.8106	0.9367	0.9360
	8	0.6034	0.6884	0.7528	0.8452	0.7647	0.8928	0.9016
	15	0.5648	0.6391	0.7267	0.8247	0.7303	0.8347	0.8683

2.5 Miscellaneous Work

There were a bunch of other smaller tasks which were required and are stated as below -

1. Prepared the conference poster and presentation by experimenting with templates and condensing the 2-page summary to fit page limits.
2. Set up the ResnetASPP environment in PyTorch, as previous TensorFlow environments were incompatible.
3. Modified U-Net to implement residual U-Net and MC Dropout, adding new predict functions to generate scaled/unscaled heatmaps and predictive-entropy plots for uncertainty experiments.
4. Performed code cleanups and backed up all results on OneDrive, including trained models, predicted images with visualizations, and the coastline and uncertainty datasets.

2.6 Future Work

We have developed the codes for the Residual and Dense U-Net and trained the models, and stored results (a qualitative analysis and uncertainty analysis remains - the framework developed here can be extended to other models as well). The future scope of this work also involves the creation of a code for U-Net++, and subsequent training of models, evaluation and testing for the ResnetASPP code and the exploration of the survey paper [19] to find any other relevant strategies or work which can be adopted. This is intended to be explored in the 1 month project to be undertaken by Sankalp.

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